



Hybrid Data Sampling for ASR: Integrating Acoustic Diversity and Transcription Uncertainty

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Abstract

Efficiently selecting training data is crucial for improving automatic speech recognition (ASR) models while minimizing annotation costs. This research extends TypiClust, a data sampling method originally validated for images, to ASR by jointly considering acoustic diversity and transcription uncertainty. Our method clusters speech embeddings from Wav2Vec2 and prioritizes typical and low transcription-confidence samples within each cluster, ensuring selection of representative and hard-to-train samples. We evaluate our method on Japanese speech datasets, CSJ and ReazonSpeech, demonstrating that it achieves lower recognition error rates than random or single-criterion-based data selection. These results indicate that selecting data based on both diversity and uncertainty enhances speech recognition model performance while reducing annotation costs.

Index Terms: automatic speech recognition, active learning, data sampling

1. Introduction

Automatic speech recognition (ASR) is essential in systems that utilize various speech technologies, such as voice dialogue systems, subtitle generation, and call center analysis [1, 2]. However, when the data distribution during training differs from that in deployment—a scenario known as domain shift—existing models often fail to perform effectively [3, 4, 5]. For example, academic conferences or television broadcasts often present unique recording conditions and specialized terminology. As a result, general-purpose speech recognition models frequently misrecognize the audio, making domain-specific models indispensable [6, 7]. Self-supervised learning models like Wav2Vec2 [8] and HuBERT [9] can improve ASR accuracy even with minimal labeled data. However, adapting these models to new speaking styles or domain-specific vocabulary still requires additional labeled data, resulting in high annotation costs.

In response to these challenges, the concept of data-centric AI has gained traction, shifting the focus from merely refining model architectures to systematically improving and selecting high-quality data [10]. In this paradigm, ensuring diverse and representative training samples is considered crucial for robust performance, especially under domain shift conditions [11]. Active learning naturally aligns with data-centric AI, as it seeks to achieve high-accuracy models within a limited annotation budget [12]. In ASR tasks, the sampling strategy must consider both acoustic diversity—across various speakers and environments—and the reliability of transcription results [13, 14, 15]. While many existing methods address either acoustic diversity or transcription uncertainty, only a limited number incorporate both in a single framework [16, 17]. Consequently, there is a need for an active learning strategy that jointly considers these

acoustic and linguistic factors.

To address the challenge of combining acoustic diversity and transcription uncertainty in ASR, our research adapts TypiClust [18], originally validated on image data, to ASR. TypiClust selects representative samples by clustering self-supervised embeddings, but its direct application to speech is non-trivial due to its temporal nature. Our method clusters acoustically similar samples using Wav2Vec2-based embeddings. It then integrates transcription confidence scores to gauge recognition reliability, allowing uncertain yet diverse data to be prioritized. This adaptation enhances ASR accuracy while reducing annotation costs.

In summary, our contributions are threefold: First, we adapts TypiClust, originally a diversity-based sampling approach for images, to speech recognition tasks. Second, the proposed method combines TypiClust’s diversity sampling with confidence-score-based uncertainty sampling, addressing both acoustic variation and transcription reliability. Third, our experiments on the CSJ and ReazonSpeech corpora—each featuring distinct acoustic properties—demonstrate our method’s effectiveness, with improvements of around 5.0 points in error rate under certain conditions compared to random sampling.

2. Related Work

2.1. Active Learning

Active learning selects samples that maximize model improvement, using either diversity-based or uncertainty-based criteria [19]. Uncertainty-based methods use entropy or confidence scores, while diversity-based methods cluster data to select representative points [20, 21]. A hybrid approach, TypiClust and Margin (TCM), has been proposed for image recognition [22], but its application to ASR remains largely unexplored.

2.2. Self-supervised learning and audio features

In recent years, self-supervised learning models such as Wav2Vec2 and HuBERT have enabled highly accurate speech feature extraction and have been widely applied to ASR tasks. High-dimensional embedding vectors from these models reportedly capture both acoustic features and some context-based linguistic information [8]. The previous reports indicate that this feature benefits clustering-based diversity sampling. In practical ASR, the confidence score has traditionally been used for active learning and for filtering transcription results [14], making it a strong indicator of the model’s uncertainty.

2.3. Active learning in speech recognition

In speech recognition, the input audio signal is a high-dimensional time series containing tens to hundreds of spectral

features per frame. Meanwhile, the output (transcribed text) is discrete and varies in length, making it challenging to align with the input [23]. As a result, active learning in speech recognition involves more complexity than in other modalities, such as images or text.

Previous studies frequently use confidence scores or prediction loss functions to concentrate annotation efforts on error-prone segments [15, 24, 25, 26]. Mainstream approaches select samples based on the acoustic model’s output probabilities and their consistency with a language model [25, 27].

Meanwhile, other research focuses on acoustic-space diversity by leveraging compact feature representations that capture speakers’ characteristics and spectral properties. For instance, certain approaches cluster Mel-Frequency Cepstral Coefficients (MFCCs)—which represent speech frequency characteristics on the Mel scale [28]—or i-vectors, which encapsulate speaker-specific details in low-dimensional vectors [16]. These clustered representations then guide active learning to select representative samples [29]. Recent work has also introduced diversity sampling methods that use high-dimensional embeddings from self-supervised learning models such as Wav2Vec2 and HuBERT [30, 31]. However, few studies simultaneously address acoustic features and transcription uncertainty, indicating a key challenge in existing research [17, 32].

Several key issues therefore arise in active learning for ASR: ensuring acoustic diversity (e.g., variations in speakers, environments, noise), managing linguistic errors (unknown words, accents, dialects), and estimating or correcting transcription uncertainty. In view of these challenges, this study integrates a clustering method (TypiClust), which uses self-supervised embeddings, with an uncertainty measure (confidence score). This integration aims to enable more effective active learning in speech recognition.

3. Proposed Method

3.1. System overview

Figure 1 shows the overall flow of the proposed method. First, Wav2Vec2 extracts utterance-level embeddings from audio samples, which are then clustered into acoustically similar groups. For each cluster, we combine a typicality measure from TypiClust with an uncertainty measure derived from the transcription confidence score to select representative samples. Originally validated on image data, TypiClust selects typical samples using density-aware scoring. We adapt this method to speech by clustering utterance-level Wav2Vec2 embeddings, increasing the likelihood of selecting acoustically diverse samples. Additionally, we integrate transcription confidence scores to assess recognition reliability and prioritize uncertain samples, improving ASR robustness under data constraints.

3.2. Ensuring diversity with TypiClust

TypiClust [18] selects the most typical sample in each cluster as the representative point. This approach has been reported to better capture local density than centroid-based sampling. It selects representative samples beyond just those near the cluster centroid, helping maintain acoustic diversity. The method then selects representative points as typical examples from these regions, prioritizing samples that best represent the acoustic and statistical properties of each cluster. Specifically, the method computes a typicality score for each sample in every cluster us-

ing the following equation.

$$\text{Typicality}(\mathbf{z}_i) = \frac{1}{\frac{1}{N} \sum_{j=1}^N d(\mathbf{z}_i, \mathbf{z}_j) + \epsilon} \quad (1)$$

Here, $d(\mathbf{z}_i, \mathbf{z}_j)$ denotes the Euclidean distance between utterance-level embeddings vectors, and N is the number of neighboring points. The term ϵ is a small constant used to maintain numerical stability. The Typicality formula sums the distances between a sample and its neighbors, where smaller distances indicate higher typicality within the cluster. Taking the reciprocal of this sum gives higher scores to more representative samples.

3.3. Uncertainty Score Estimation via Confidence Score

Our method computes a confidence score based on the ASR model’s transcription results. It then prioritizes annotation for utterances with high uncertainty that is likely to misrecognize. The proposed method uses the confidence score of transcribed text derived from an existing ASR model. The confidence score is defined as the maximum softmax probability of the predicted word in the ASR model’s output. ASR model’s softmax probabilities were adopted as confidence scores for their reliability in estimating transcription accuracy. The following equation computes the uncertainty score.

$$\text{Uncertainty Score}(x_i) = 1 - \max_{y \in Y} P(y | x_i) \quad (2)$$

Here, $P(y | x_i)$ represents the probability that the ASR model predicts text y given the acoustic feature x_i . This approach employs the score to identify samples with high uncertainty. It then prioritizes these samples for selection within each cluster.

3.4. Integrating Diversity and Uncertainty

Before combining TypiClust with the confidence score, we normalize TypiClust’s typicality values to the range $[0, 1]$. The method then quantifies clustering-based diversity using this normalized typicality score and measures uncertainty using the confidence score.

- **Clustering:** Wav2Vec2 utterance-level embeddings are clustered into k groups.
- **In-Cluster Sampling:** The method calculates an integrated score for each sample (as described below) and selects the top m samples per cluster.
- **Adjusting the Sampling Ratio:** Users can adjust the balance between uncertainty and diversity to fit the task.

The final integrated score is computed as follows:

$$S(x_i, \mathbf{z}_i) = \lambda \cdot \text{Typicality}(\mathbf{z}_i) + (1 - \lambda) \cdot \text{Uncertainty Score}(x_i) \quad (3)$$

Here, λ is a hyperparameter that balances diversity and uncertainty. We test $\lambda = 0.3$ (uncertainty-focused), 0.5 (balanced), and 0.7 (diversity-focused) to analyze how varying λ affects performance across various data types and volumes. Algorithm 1 shows the details of the flow of the previous method.

4. Experimental Setup

4.1. Dataset

This study used two Japanese speech corpora for evaluation: CSJ¹ and ReazonSpeech². We provide an overview of each cor-

¹<https://clrd.ninjal.ac.jp/csj/>

²<https://research.reazon.jp/projects/ReazonSpeech/index.html>

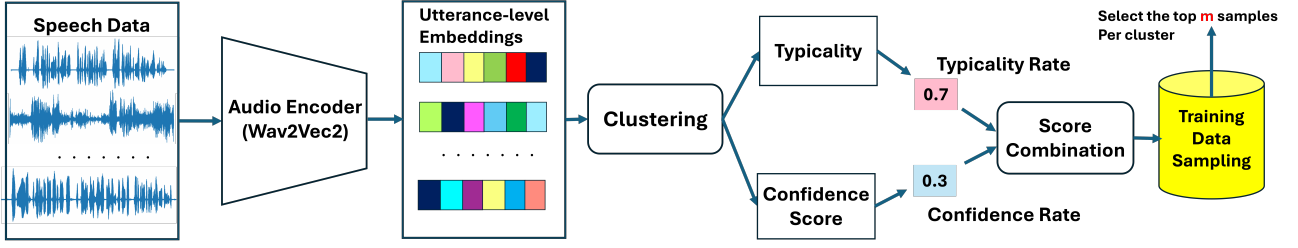


Figure 1: Overview of proposed method

Algorithm 1 Extending TypiClust for Speech Recognition Data Sampling Based on Confidence Score

Require: Speech dataset: $D = \{x_i\}$,
 Pretrained embedding extractor: $E(\cdot)$,
 Number of clusters: K ,
 Minimum cluster size: minSize ,
 Samples per cluster: m ,
 Weight parameter: λ

Ensure: Sampled training dataset: S

```

1: Step 1: Embedding Extraction
2: for  $x_i \in D$  do
3:    $z_i \leftarrow E(x_i)$ 
4: end for
5:  $Z \leftarrow \{z_i \mid x_i \in D\}$ 
6: Step 2: Clustering
7:  $\text{clusters} \leftarrow \text{kmeans}(Z, K)$ 
8: Step 3: Score Calculation and Sample Selection
9: for  $c = 1$  to  $K$  do
10:   $I_c \leftarrow \{i \mid \text{clusters}[i] = c\}$ 
11:  if  $|I_c| < \text{minSize}$  then
12:    skip // (Excluding small clusters.)
13:  else
14:    for  $i \in I_c$  do
15:       $t_i \leftarrow \text{CalcTypicality}(z_i, Z_{I_c})$ 
16:       $u_i \leftarrow \text{CalcUncertaintyScore}(x_i)$ 
17:       $s_i \leftarrow \lambda t_i + (1 - \lambda) u_i$ 
18:    end for
19:     $\text{sort}(I_c, \text{by descending } s_i)$ 
20:     $S_c \leftarrow \text{top}(I_c, m)$ 
21:  end if
22: end for
23: Step 4: Integration of Selected Results
24:  $S \leftarrow \bigcup_{c=1}^K S_c$ 
25: return  $S$ 

```

pus below.

- CSJ: Japanese speech dataset covering presentations and conversations with diverse speakers and environments.
- ReazonSpeech: Japanese speech corpus recorded exclusively from television broadcasts, encompassing diverse speakers and recording conditions.

In the experimental procedure, 10 hours of data from each dataset were set aside for testing and excluded it from the training set. Then, the proposed algorithm sampled 10, 50, 100 hours of training data.

Proposed Method (TypiClust + Confidence Score)

- The method applies K-means clustering to the 1024-

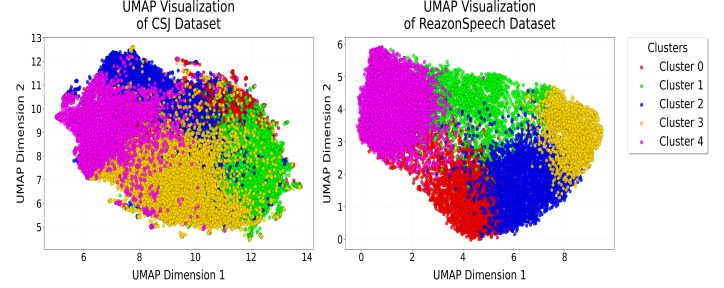


Figure 2: Cluster visualization of Wav2Vec2-based acoustic features for both the CSJ and ReazonSpeech corpora.

dimensional embedding vectors extracted by Wav2Vec2-Large³.

- In each cluster, the approach calculates a typicality score via TypiClust. It also accounts for the uncertainty derived from the confidence score of the initial, non-fine-tuned OpenAI/Whisper Medium⁴ [19].
- The approach selects the top m samples with the highest diversity-uncertainty scores, setting m to match the predefined training set duration.
- The number of clusters is set to $K = 5$ in accordance with the elbow method. The typicality score uses $N = 10$ neighbors for each data sample.

Figure 2 shows the two-dimensional UMAP projection of the CSJ and the ReazonSpeech audio embedding vectors obtained through Wav2Vec2.

Comparison method

For comparison, the study also assessed Random Sampling. In addition, ablation studies were conducted using only TypiClust or only the confidence score.

4.2. Speech Recognition Model and Training Conditions

This study uses OpenAI/Whisper Medium as the base speech recognition model. This model contains approximately 769 million parameters, making it relatively large, and has gained attention as a multilingual baseline model. In our experiments, we used PyTorch and Hugging Face Transformers for implementation. The training employed a batch size of 8, a learning rate of 1.0×10^{-6} , and a total of 5 epochs. All training was conducted on a single NVIDIA A100 GPU with 80 GB of memory.

³<https://huggingface.co/ljvdata/wav2vec2-large-xlsr-53-japanese>

⁴<https://github.com/openai/whisper.git>

Table 1: *Experimental Results*

Dataset(h)	Method	CSJ		ReazonSpeech	
		CER (%)	WER (%)	CER (%)	WER (%)
10h	Random Sampling	30.21 $\pm$ 0.62	35.41 $\pm$ 0.69	29.89 $\pm$ 0.64	23.86 $\pm$ 0.80
	TypiClust only	28.49 $\pm$ 0.64	34.82 $\pm$ 0.71	26.39 $\pm$ 0.72	22.94 $\pm$ 0.72
	confidence score only	27.32 $\pm$ 0.60	33.81 $\pm$ 0.70	28.46 $\pm$ 0.72	24.68 $\pm$ 0.73
	TypiClust+confidence score ($\lambda = 0.3$)	25.64 $\pm$ 0.62	31.64 $\pm$ 0.70	25.98 $\pm$ 1.04	23.50 $\pm$ 1.11
	TypiClust+confidence score ($\lambda = 0.5$)	26.44 $\pm$ 0.69	32.33 $\pm$ 0.75	26.13 $\pm$ 1.04	23.35 $\pm$ 1.11
	TypiClust+confidence score ($\lambda = 0.7$)	26.52 $\pm$ 0.66	32.90 $\pm$ 0.74	26.52 $\pm$ 0.74	21.90 $\pm$ 0.73
50h	Random Sampling	26.47 $\pm$ 0.67	32.23 $\pm$ 0.73	25.21 $\pm$ 0.66	21.59 $\pm$ 0.74
	TypiClust only	29.63 $\pm$ 0.71	35.82 $\pm$ 0.75	21.12 $\pm$ 0.71	17.27 $\pm$ 0.70
	confidence score only	22.08 $\pm$ 0.60	27.47 $\pm$ 0.68	23.86 $\pm$ 1.31	18.22 $\pm$ 1.24
	TypiClust+confidence score ($\lambda = 0.3$)	25.72 $\pm$ 0.55	31.61 $\pm$ 0.64	21.27 $\pm$ 0.72	17.89 $\pm$ 0.73
	TypiClust+confidence score ($\lambda = 0.5$)	23.35 $\pm$ 0.49	29.07 $\pm$ 0.59	21.02 $\pm$ 0.71	17.46 $\pm$ 0.72
	TypiClust+confidence score ($\lambda = 0.7$)	21.50 $\pm$ 0.56	26.54 $\pm$ 0.64	23.18 $\pm$ 0.72	18.04 $\pm$ 0.72
100h	Random Sampling	25.60 $\pm$ 0.77	31.23 $\pm$ 0.82	20.30 $\pm$ 0.70	17.29 $\pm$ 0.72
	TypiClust only	29.45 $\pm$ 0.63	35.42 $\pm$ 0.70	19.14 $\pm$ 0.69	16.94 $\pm$ 0.70
	confidence score only	21.01 $\pm$ 0.57	25.96 $\pm$ 0.65	20.80 $\pm$ 0.70	17.45 $\pm$ 0.72
	TypiClust+confidence score ($\lambda = 0.3$)	20.90 $\pm$ 0.57	26.05 $\pm$ 0.66	20.22 $\pm$ 0.68	16.98 $\pm$ 0.69
	TypiClust+confidence score ($\lambda = 0.5$)	20.77 $\pm$ 0.48	26.08 $\pm$ 0.60	20.43 $\pm$ 0.68	17.01 $\pm$ 0.69
	TypiClust+confidence score ($\lambda = 0.7$)	20.78 $\pm$ 0.51	25.80 $\pm$ 0.61	19.41 $\pm$ 0.69	17.13 $\pm$ 0.70

OpenAI/Whisper Medium was chosen as the fine-tuning model because it is an end-to-end ASR system and natively provides transcription confidence scores, making it well-suited for uncertainty-based sampling.

4.3. Evaluation Metrics

This study evaluates ASR performance using Character Error Rate (CER) and Word Error Rate (WER). These two metrics measure character-level and word-level recognition errors, respectively.

5. Experimental results and discussion

5.1. Results

Tables 1 show the experimental results on the CSJ and ReazonSpeech corpora, respectively. Each numerical value indicates an error rate alongside its 95% confidence interval. In most cases, the proposed method (TypiClust + confidence score) achieved lower CER and WER. On the CSJ corpus, it improved CER by approximately 4.79 points and WER by about 5.29 points compared to Random Sampling. On ReazonSpeech, it improved CER by around 3.37 points and WER by 2.76 points over Random Sampling. Although using TypiClust or the confidence score alone still yielded some improvements, jointly considering both diversity and uncertainty generally led to greater performance gains.

5.2. Discussion

The proposed method generally outperforms random sampling and single-criterion approaches with less data, making it well-suited for resource-constrained scenarios. Performance improves with more training data, as observed in the results for 10, 50, and 100 hours. While TypiClust + confidence score performs best for CSJ, using only TypiClust yields better results on ReazonSpeech when the training audio exceeds 50 hours. This advantage is likely due to ReazonSpeech’s extensive acoustic diversity.

Lower λ values improved performance with 10 hours of

data, while higher values were more effective with 50 or 100 hours. This pattern suggests that when the sampling duration is short, uncertainty contributes more to performance, but as sampling time increases, acoustic diversity becomes a stronger determinant. Interestingly, this trend differs from findings in prior image-based research [22], indicating a need for future work to examine the distinct data characteristics of image and audio domains.

In practical deployments, a straightforward way to choose λ is to evaluate multiple λ settings on a validation set drawn from the target domain and then select the value that yields the lowest error rate. This strategy ensures that the chosen sampling balance aligns best with the dataset’s characteristics.

6. Conclusion

This study proposes a hybrid sampling method for active learning in ASR. It combines clustering-based typical sample selection (TypiClust) with a confidence score. By clustering Wav2Vec2 utterance-level embeddings and prioritizing high-uncertainty samples while preserving diversity, the method contributes to ASR performance improvement while reducing annotation costs.

Experiments on the CSJ and ReazonSpeech corpora demonstrate that the proposed approach generally outperforms random sampling and single-criterion strategies in CER and WER, though TypiClust alone performs better in some cases.

In future work, we plan to explore uncertainty estimation directly from self-supervised models and develop dynamic clustering strategies that automatically adjust cluster counts and sample selection. Additionally, we will compare our method with state-of-the-art sampling approaches and evaluate its effectiveness on languages beyond Japanese. We will also address practical considerations such as inference time, memory overhead, and computational cost associated with embedding extraction and clustering, aiming to enhance efficiency for large-scale ASR applications. Addressing these challenges will enable more efficient and scalable active learning strategies for ASR, further enhancing real-world applicability.

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8. References

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